

Cable simulation: which approach to adopt

A decision brief — compatibility first

The question. A cable must be simulated as part of a larger system. Several methods reproduce its hanging shape to within a few centimetres of a real cable, so raw accuracy is *not* the deciding factor. What decides it is **what each approach is compatible with** — which existing stack it plugs into with the least friction — because the cable is one component of a much bigger project.

One thing to know about accuracy. The method closest to the textbook mathematics is the *furthest* from the real cable, and vice-versa. This is not a solver being wrong: the real cable is held with tape that fixes its angle, which the ideal curve does not model. Every method here is behaving correctly; they differ in *which* real-world detail they capture. So choose on integration, and treat all of them as accurate enough to a few centimetres.

Approach	vs real	speed	Integrates with
PhysX capsule chain	33 mm	29 s	Drops straight into an Isaac Sim / Omniverse project: USD-native, works with Isaac Lab, the ROS 2 bridge, sensors, and collision against robots and the environment. Real-time.
<i>Best when the cable must live in, and interact with, an Isaac Sim scene. Watch: rigid-capsule approximation; not differentiable.</i>			
Newton cable (add_rod)	75 mm	13 s	Native to the Newton / Warp stack: differentiable (gradients for control, learning, system identification) and GPU-batched for large-scale training. Newton is the forward-looking successor in the Isaac roadmap.
<i>Best when high-fidelity cable physics, differentiability, or GPU-batched learning are needed. Watch: newer, evolving API; heavier per-step cost.</i>			
Warp rod	51 mm	1 s	A single self-contained Warp component with no engine dependency: differentiable, portable, easy to embed in a custom pipeline. You own the modelling (stiffness, resolution, boundary conditions).
<i>Best when a lightweight or differentiable rod is wanted without a full engine. Watch: bending stiffness and mesh resolution must be calibrated; crudest bending model.</i>			

Recommendation.

- If the wider project lives in **Isaac Sim / Omniverse** and the cable must interact with robots or the scene: **PhysX capsule chain**. Most compatible with what is already there, closest to the real cable, and real-time.
- If the project needs **differentiability or GPU-batched learning**, or is adopting **Newton: Newton cable**. Highest fidelity to the physics and future-facing, at a higher compute cost.
- If a **lightweight, embeddable, engine-free** component is wanted: **Warp rod**. Fewest dependencies and differentiable, but you calibrate its stiffness and resolution.

Ruled out on this build. A Newton *USD-engine* cable cannot be fixed at both ends (an articulation must be a tree, a doubly-clamped cable is a loop), and PhysX *volumetric FEM* builds but its state cannot be read back. Both fail with a clear diagnosis rather than silently. The three above are the live options.

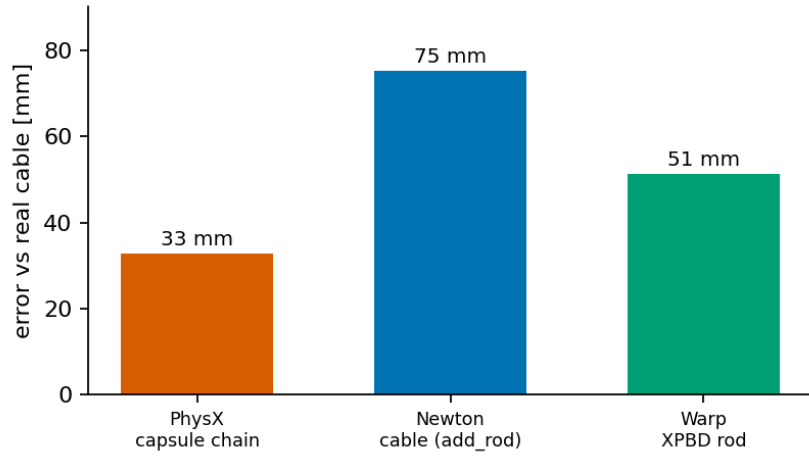


Figure 1: Error against the photographed cable (lower is closer to reality).

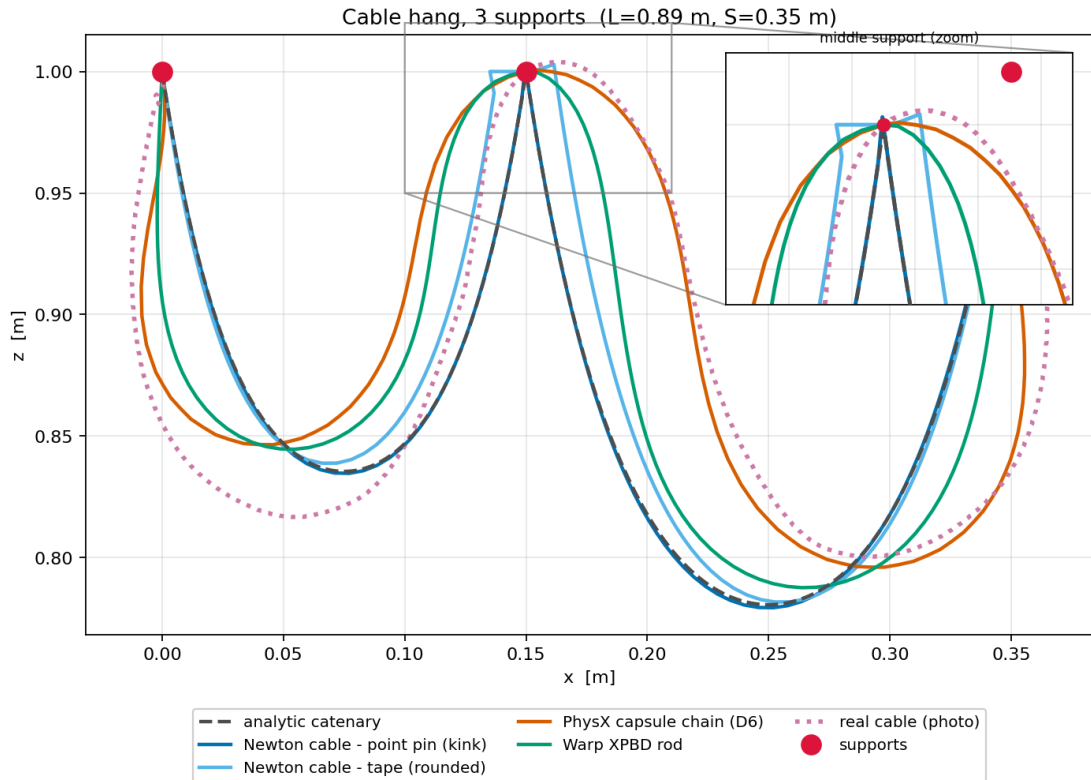


Figure 2: Every approach, the exact mathematics (dashed), and the real cable (dotted) on one axis. The methods bracket the real cable; the differences are boundary-condition modelling, not solver bugs.